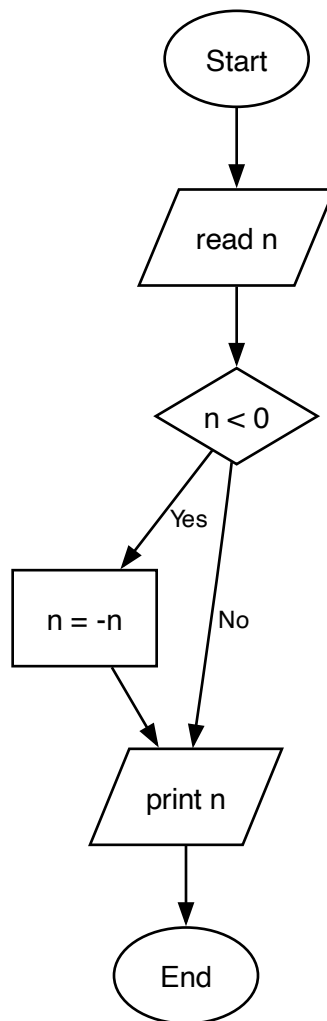


Problem 1.



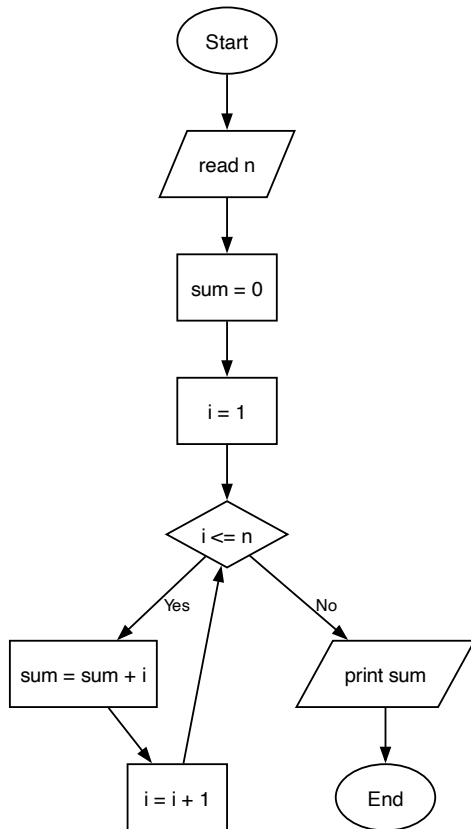
(i) For each input below, follow the flowchart and write the output.

input n	output
7	
-4	
0	
-19	

(ii) Can you tell what this algorithm is doing?

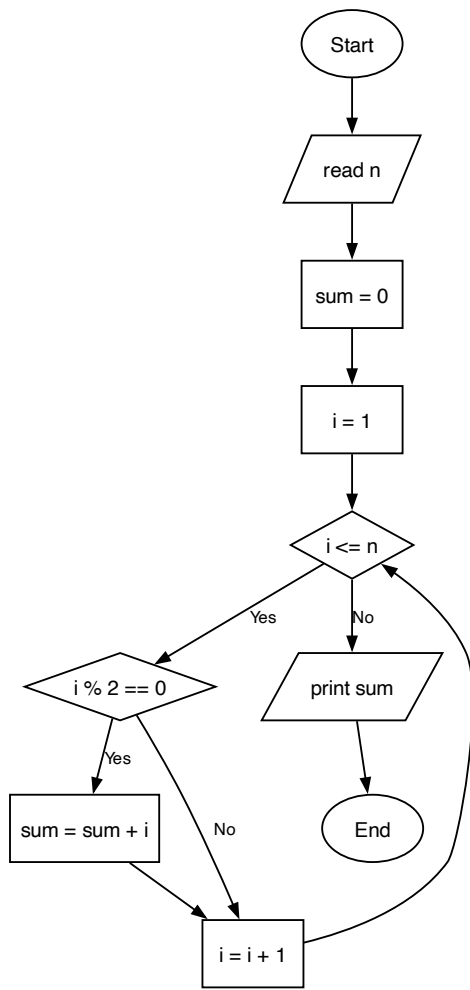
Problem 2.

Trace this algorithm for $n = 4$, and complete the trace table.

[illegible]

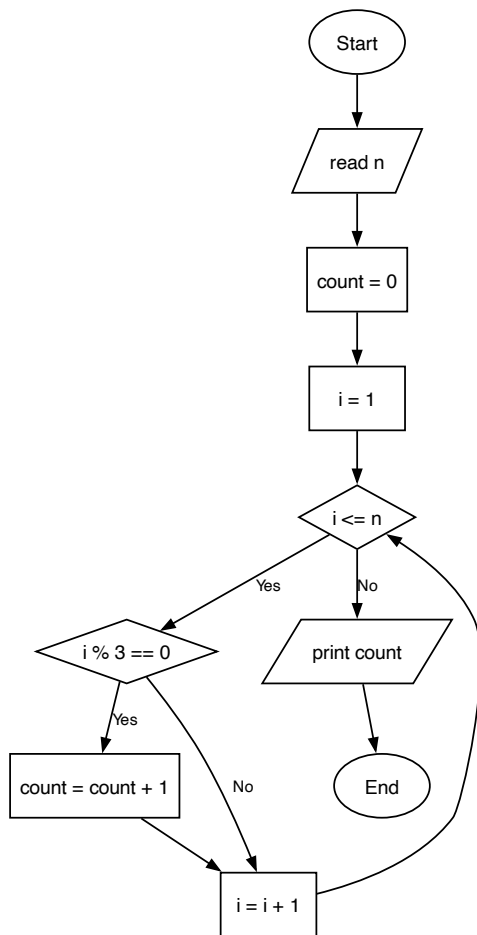
Problem 3.

Trace this algorithm for $n = 5$, and complete the trace table.

[illegible]

Problem 4.

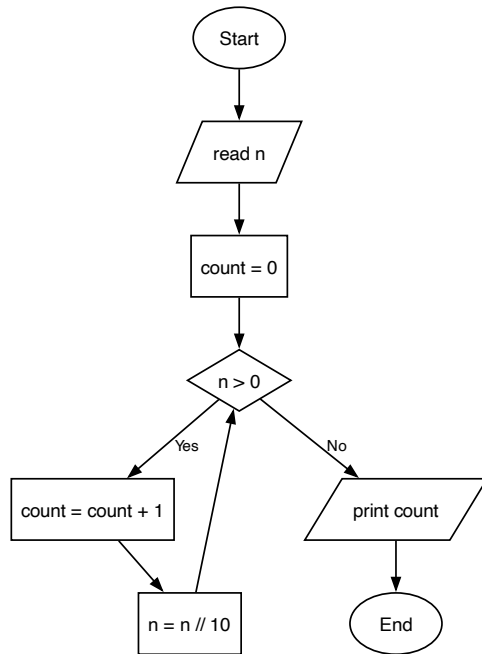
Trace this algorithm for $n = 7$, and complete the trace table.

[illegible]

Problem 5.

Trace this algorithm for $n = 472$, and complete the trace table.

Note: $n // 10$ means whole-number division — divide and drop the remainder. For example, $472 // 10 = 47$ (not 47.2).

[illegible]

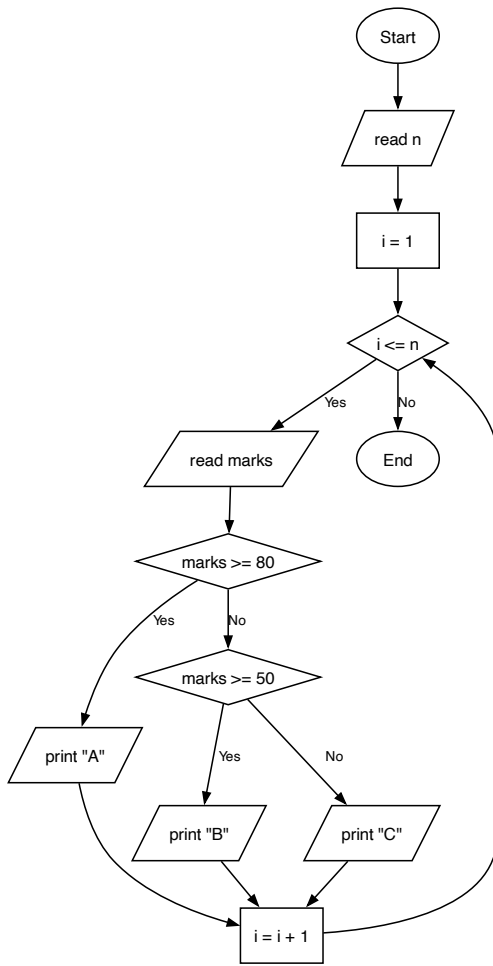
Problem 6.

Draw a flowchart for this algorithm:

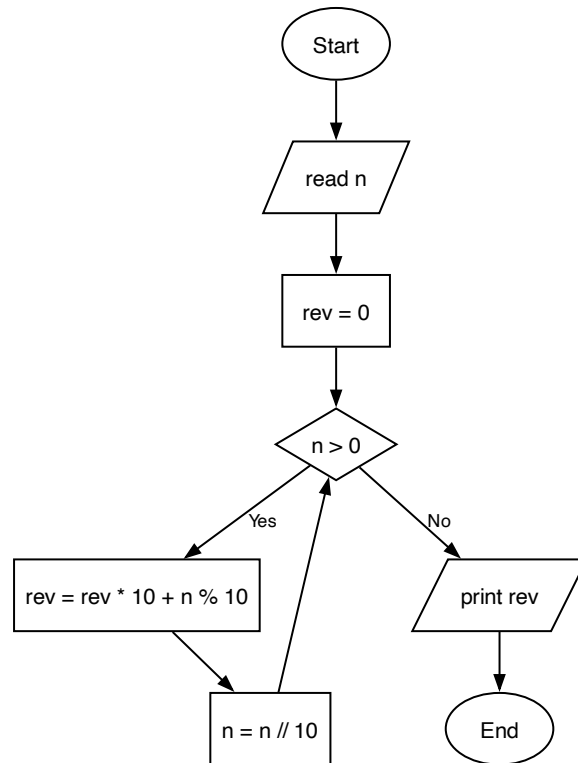
Ask the user for a number n . Then go through the numbers 1, 2, 3, ... up to n , one by one: if a number is divisible by 3, print “buzz”; otherwise print the number itself.

Problem 7.

Trace this algorithm for $n = 3$, where the values of *marks* entered are 55, 82, 40 (in that order), and complete the trace table.

[illegible]

Problem 8.



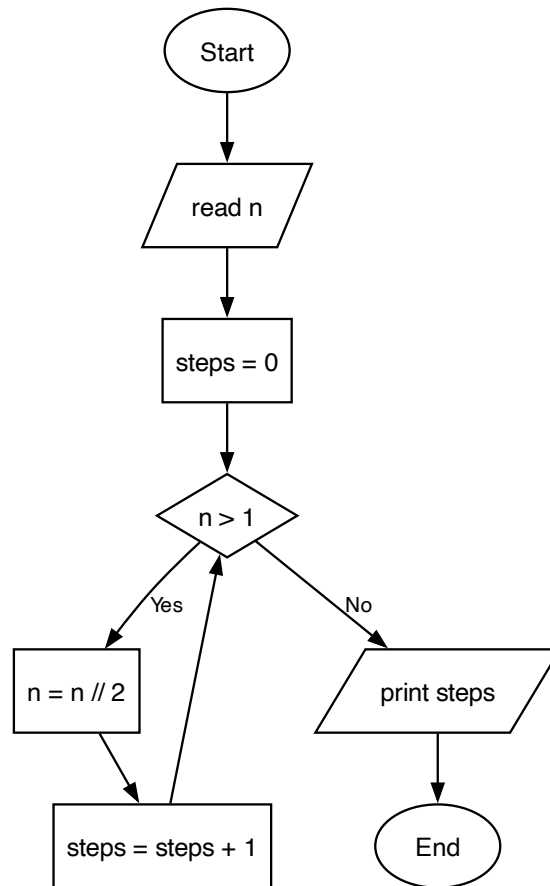
(i) For each input below, follow the flowchart and write the output.

Note: $n \% 10$ gives the remainder after dividing by 10 — the last digit of n . For example, $472 \% 10 = 2$.

input n	output
123	
47	
5	
470	

(ii) Can you tell what this algorithm is doing?

Problem 9.



(i) For each input below, follow the flowchart and write the output.

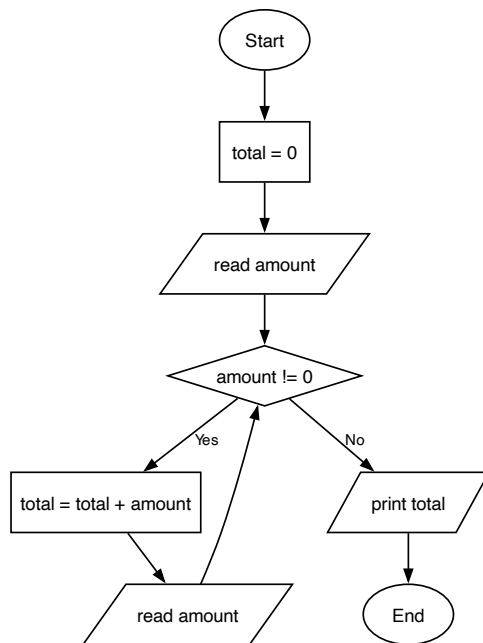
input n	output
8	
16	
100	
1000	

(ii) n grew from 8 all the way to 1000 — but look how little the output grew! Can you explain why?

Problem 10.

A shopkeeper adds up the money received from each customer at the end of the day. She enters the amounts one by one, and enters 0 to say “no more” — the till then shows the total. Notice that the algorithm cannot know in advance how many customers there were!

Trace this algorithm, where the values of *amount* entered are 120, 75, 200, 0 (in that order), and complete the trace table.



total	amount	output